Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 4+ years deploying enterprise-scale ML and GenAI solutions into production. Expert in building end-to-end MLOps pipelines with model governance, drift detection, and automated retraining on AWS. Proven track record of working in diverse multi-national teams and taking AI models from experimentation to production, processing sensitive social sciences datasets with 95%+ accuracy while ensuring scalable deployment.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Architected production-grade CI/CD pipeline for AI-powered development workflow using GitHub Actions, and Vercel, enabling automated deployment of codebases with 90%+ code generation rate and weekly release cycles.
- Designed automated testing and quality assurance framework adapted for AI-generated code, implementing validation pipelines that reduced manual code review time by 70% while maintaining production stability and security standards.
- Developed end-to-end MLOps infrastructure for multi-tenant SaaS platform serving global B2B companies, accelerating feature delivery from monthly to weekly releases through AI-driven development automation.
- Deployed the Java application using Maven-framework for a gearswitch company to digitise the process of reporting of the installation asset examination with integrated LLM support for summarisation of the report content.

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific GenAI, scikit-learn and PyTorch models, and model hierarchies using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment, achieving 95%+ accuracy on sensitive social science datasets.
- Designed serverless ML infrastructure on AWS (Lambda, Step Functions, EFS) with automated model retraining pipelines, reducing manual data validation workload by 50% and enabling non-technical staff to process complex datasets independently.
- Optimized statistical disclosure control algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement and enabling real-time processing of datasets with 500k+ records.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK and the USA.
- Taught Data Structures, C/C++, and ML to 100+ undergrad/postgrad students.

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Built a Python-based Raspberry Pi self-hosted Ollama MCP server with Twitch chat commands integration to respond to chat prompts during stream.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a real-time trading simulation chatbot in Go integrated with Twitch chat, with a backend API recording stock purchases in PostgreSQL (Supabase).
- Developed a JavaScript UI hosted online to visualize and track portfolio performance in real time with interactive graphs and table.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Deployed the Golang REST API with LLM integration to map the card data against the PostgreSQL database.
- Developed React application deployed on vercel for the project frontend.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Developed PyTorch-based transfer learning framework for Bayesian model averaging in financial forecasting; published peer-reviewed papers and released open-source package adopted externally.
- Listed as the main author on the internationally acknowledged scientific journal publication and co-authored 3 scientific conference papers.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Built C++ agent-based financial simulation modeling macroeconomic dynamics under negative interest rates.